

- 4 Obesity indicators are used to assess the risk of cardiovascular disease (CVD), including coronary heart disease (CHD).

Obesity indicators include body mass index (BMI) and waist-to-hip ratio (WHR).

- (a) Describe the differences between CVD and CHD.

(2)

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- (b) A woman who was 165 cm tall had a mass of 68 kg, a waist circumference of 85 cm and a hip measurement of 101 cm.

- (i) Calculate the BMI of this woman.

Use the formula:

$$\text{BMI} = \frac{\text{mass in kg}}{(\text{height in m})^2}$$

(1)

Answer

- (ii) Calculate the WHR of this woman.

(1)

Answer : 1

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- (c) Scientists collected data to evaluate these two obesity indicators for their effectiveness in predicting death from CVD and CHD.

This study involved 9 206 adults aged from 20 to 69 years from Australian cities.

This study started in 1989, when measurements for the obesity indicators were taken.

Risk factors including blood pressure and blood cholesterol levels were measured.

Information about other risk factors was recorded, but not measured.

By the year 2000, 473 of these adults had died. The data were compared with the causes of death.

- (i) Explain why high blood pressure is a risk factor for CVD.

(2)

- (ii) Some other risk factors were recorded as being either present or absent but were not measured.

Name **two** other risk factors that might have been recorded but not measured.

(1)

1

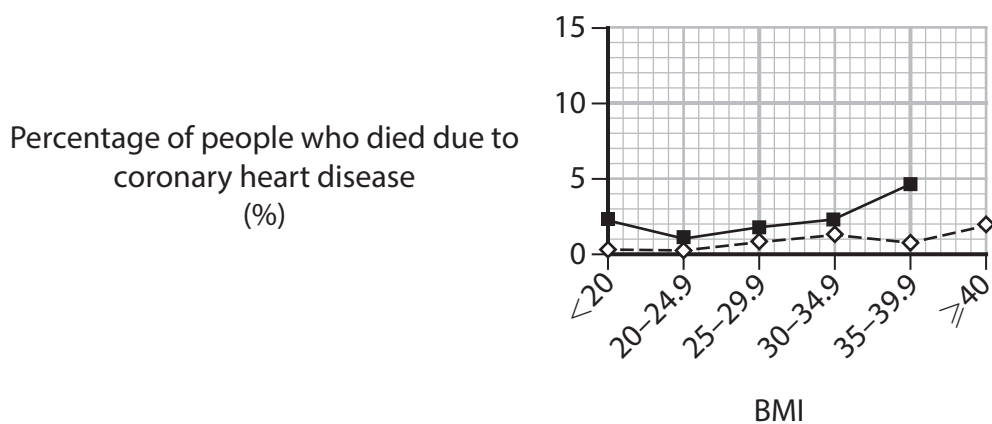
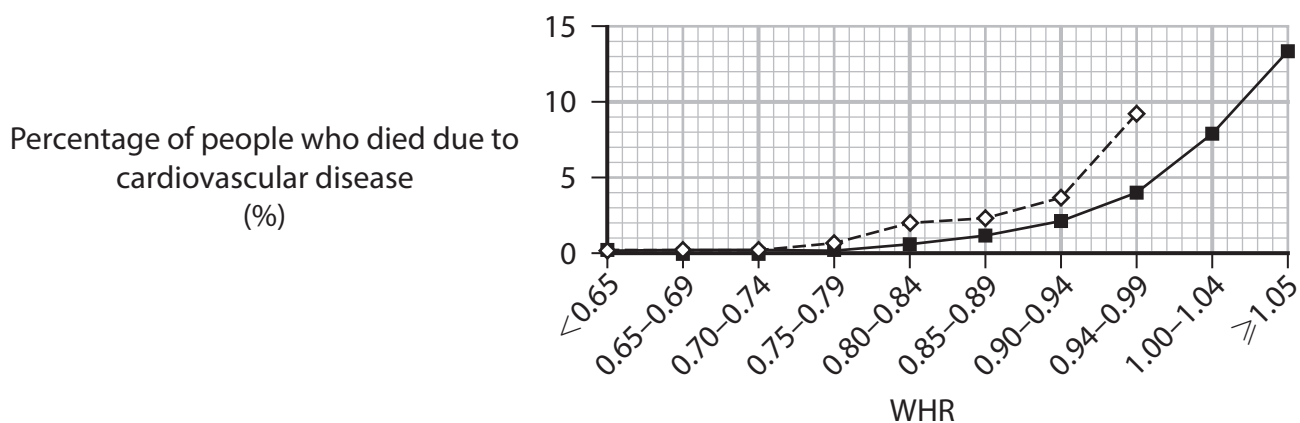
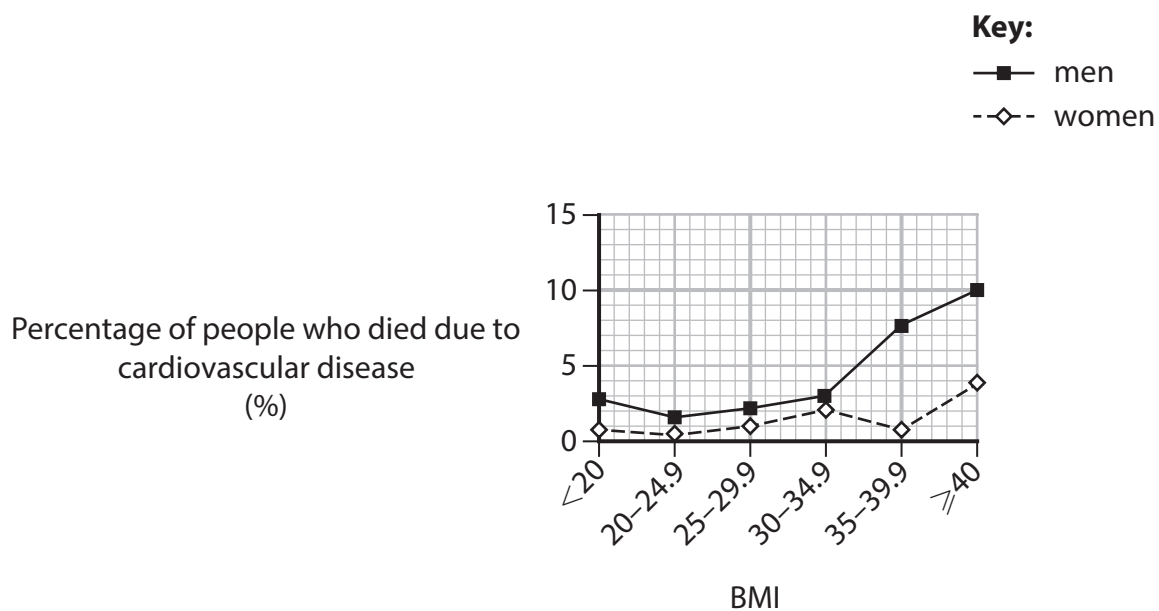
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- (iii) The graphs show the BMI and WHR of the men and women who died due to either CVD or CHD.

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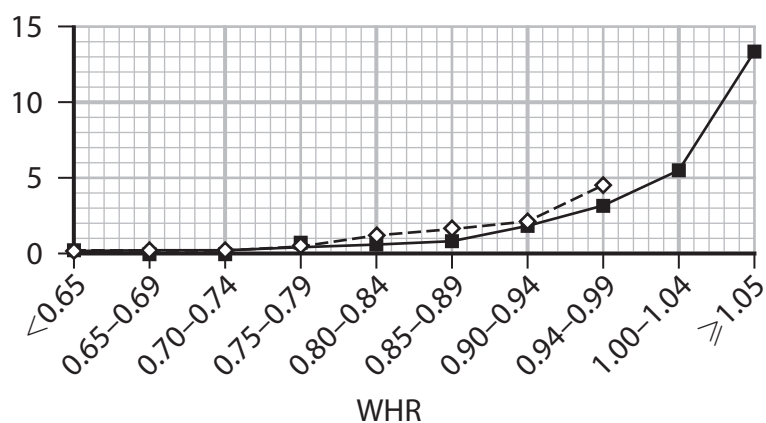


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Percentage of people who died due to
coronary heart disease
(%)



Discuss the usefulness of these two obesity indicators in predicting death from CVD and CHD.

(3)

(Total for Question 4 = 10 marks)